

WATONGA, OK, USA

WMO: 720342

Lat: 35.864N

Lon: 98.421W

Elev: 1548

StdP: 13.89

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.2	17.8	4.2	7.2	15.2	9.2	9.3	21.0	29.5	52.2	27.1	45.4	9.4	30	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	99.2	75.6	96.9	75.4	94.4	75.2	78.7	91.2	77.8	90.4	77.0	89.4	11.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.7	142.7	82.9	74.8	138.4	82.3	73.5	132.0	81.3	43.5	91.4	42.5	91.3	41.6	89.2	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.3	23.8	20.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.3	38.6	40.9	52.1	58.2	68.3	79.0	81.3	80.7	74.1	61.2	49.3	38.9
	DBStd	17.97	10.25	11.44	10.55	9.62	8.16	5.38	5.43	5.24	6.74	9.77	9.90	9.56
	HDD50	1308	372	291	102	35	2	0	0	0	0	20	128	358
	HDD65	3697	818	675	412	244	62	0	0	0	11	190	477	808
	CDD50	5078	19	37	167	280	569	870	971	951	723	368	107	16
	CDD65	1991	0	1	12	39	164	420	507	486	284	73	5	0
	CDH74	22047	1	19	112	421	1560	4820	6159	5520	2741	646	47	1
	CDH80	10583	0	4	17	111	577	2376	3315	2845	1152	180	6	0
Wind	WSAvg	9.4	9.8	10.2	11.1	11.6	9.4	10.0	8.0	7.3	7.7	9.0	9.9	9.3
Precipitation	PrecAvg	30.9	1.0	1.3	2.5	3.0	4.0	4.1	2.4	3.9	2.5	2.9	1.9	1.4
	PrecMax	54.3	2.7	4.1	7.2	7.9	9.6	13.7	5.7	8.1	5.6	7.5	5.8	4.0
	PrecMin	20.1	0.0	0.0	0.0	0.3	0.7	0.4	0.4	0.0	0.0	0.0	0.0	0.1
	PrecStd	7.7	0.8	1.1	2.0	1.8	2.1	3.1	1.3	2.4	1.6	2.0	1.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.3	76.8	81.9	89.6	94.9	99.8	104.4	101.5	96.2	89.6	80.0	68.5
		MCWB	55.3	59.1	61.5	66.9	71.5	74.9	76.2	76.1	73.8	70.3	64.5	56.5
	2%	DB	65.1	71.1	77.0	82.4	90.4	96.9	99.1	99.2	92.8	84.8	73.8	63.8
		MCWB	51.6	56.1	61.8	66.1	72.0	75.2	76.1	75.7	74.2	70.6	60.9	55.3
	5%	DB	60.6	64.2	72.9	78.9	87.0	93.7	96.8	96.7	90.3	81.1	69.7	58.9
		MCWB	48.8	53.8	61.1	63.9	71.4	74.4	76.0	75.6	73.6	67.7	59.5	50.2
	10%	DB	55.2	59.3	69.4	74.9	82.3	91.2	94.1	93.2	87.6	77.2	65.1	54.8
		MCWB	46.1	50.6	59.7	62.7	70.2	74.2	75.8	75.3	72.7	65.7	57.6	47.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	62.6	68.9	72.2	77.4	80.2	80.2	79.6	78.3	75.3	68.3	63.4
		MCDB	65.7	68.6	74.8	79.4	86.7	91.6	92.9	92.8	90.2	82.4	74.5	64.8
	2%	WB	53.7	58.9	66.8	69.6	75.7	78.3	78.8	78.2	76.6	73.3	64.3	56.1
		MCDB	63.2	66.7	72.7	78.3	84.3	90.4	91.4	91.2	87.5	80.3	68.4	60.8
	5%	WB	50.1	55.4	64.3	67.5	74.0	77.2	77.9	77.3	75.3	71.2	61.2	51.1
		MCDB	58.2	65.4	69.0	75.6	82.4	89.3	91.0	90.0	85.1	77.6	66.7	57.7
	10%	WB	46.8	50.9	61.2	65.3	72.0	76.1	77.0	76.3	74.0	68.3	58.4	47.4
		MCDB	54.5	58.4	67.1	72.3	79.6	87.6	90.1	88.8	83.1	76.0	64.7	54.3
Mean Daily Temperature Range	5% DB	MDBR	21.8	22.0	23.3	23.6	20.9	20.9	21.4	20.9	20.6	22.0	22.1	20.2
		MCDBR	29.4	29.8	26.4	27.7	25.2	23.4	24.4	23.9	22.7	24.3	27.0	27.0
		MCWBR	18.4	17.9	15.6	15.1	11.5	8.1	7.2	7.3	8.1	12.3	16.7	17.7
	5% WB	MCDBR	27.0	27.8	21.0	23.8	21.0	20.8	21.4	21.4	19.6	19.8	23.4	24.6
		MCWBR	17.7	18.2	14.4	15.0	11.2	9.0	7.7	7.9	8.3	11.5	16.8	18.3
Clear-Sky Solar Irradiance	taub	0.282	0.291	0.333	0.370	0.398	0.415	0.414	0.417	0.375	0.343	0.303	0.281	
	taud	2.522	2.509	2.407	2.333	2.327	2.321	2.342	2.286	2.379	2.454	2.505	2.526	
	Ebn at Noon	295	304	297	290	281	275	275	272	279	280	283	288	
	Edh at Noon	25	29	35	40	41	41	40	41	36	30	25	23	
All-Sky Solar Radiation	RadAvg	900	1102	1455	1801	1972	2208	2202	1985	1659	1249	948	762	
	RadStd	86	157	102	134	199	179	123	117	117	161	122	80	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page